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|---|---|
| **Started on** | Wednesday, 30 September 2026, 10:28 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 30 September 2026, 10:29 PM |
| **Time taken** | 38 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

## Question 1

Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>

void quickSort(int a[], int l, int h)
{
    if (l >= h)
        return;

    int i = l, j = h;
    int p = a[(l + h) / 2];

    while (i <= j)
    {
        while (a[i] < p)
            i++;

        while (a[j] > p)
            j--;

        if (i <= j)
        {
            int t = a[i];
            a[i] = a[j];
            a[j] = t;
            i++;
            j--;
        }
    }
```

```
    quickSort(a, l, j);
    quickSort(a, i, h);
}

int main()
{
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    quickSort(a, 0, n - 1);

    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

**Correct**

Marks for this submission: 1.00/1.00.